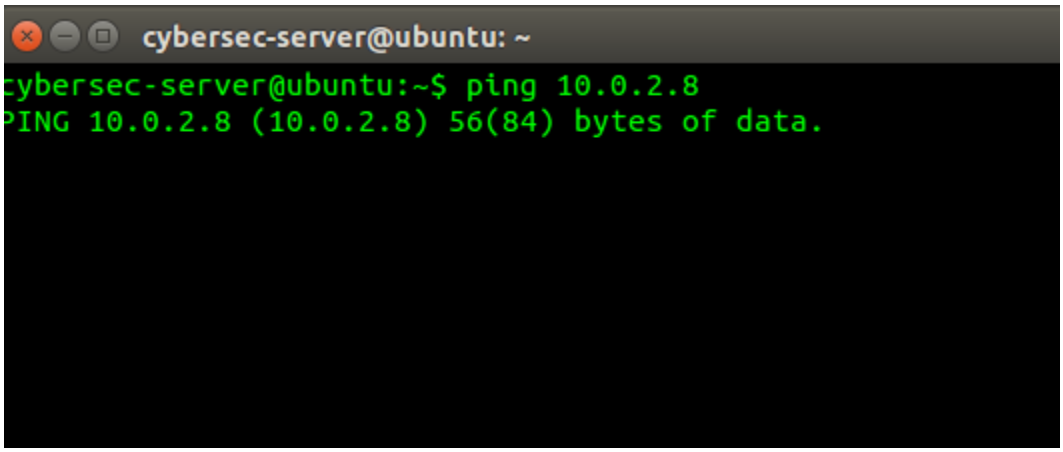
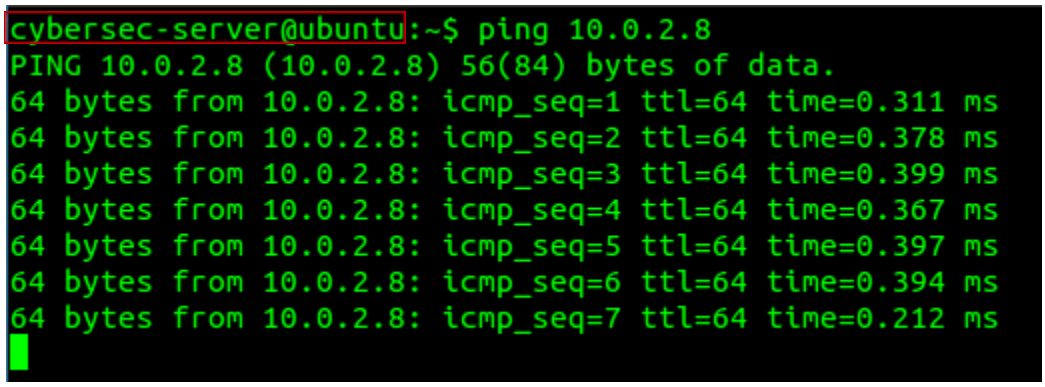
CYBERSECURITY LAB 8

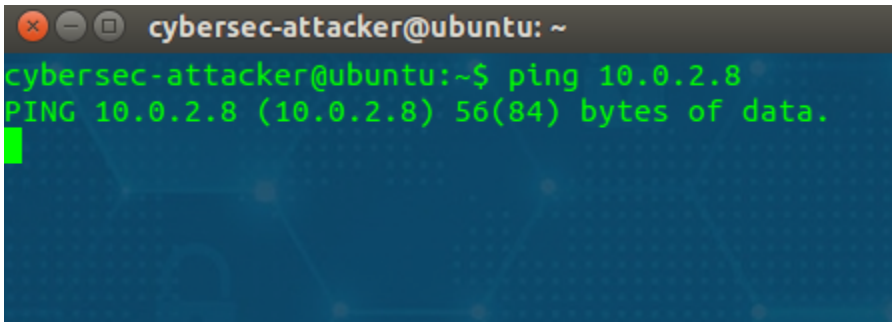
Firewall Configuration with iptables



The above image describes the scenario when the rules are set to drop all traffic.



The above image describes the scenario where the rules were set to accept traffic from 10.0.2.6 (cybersec-server) and hence the ping was successful.



Continuing from the same above scenario the ping is unsuccessful from the attacker vm.

```
cybersec-attacker@ubuntu: ~
cybersec-attacker@ubuntu:~$ ssh cybersec-client@10.0.2.8
```

Output for ssh connection before configuring the rules.

```
cybersec-client@ubuntu: ~
cybersec-attacker@ubuntu:~$ ssh cybersec-client@10.0.2.8
^Z
[1]+  Stopped                 ssh cybersec-client@10.0.2.8
cybersec-attacker@ubuntu:~$ ssh cybersec-client@10.0.2.8
cybersec-client@10.0.2.8's password:
Welcome to Ubuntu 14.04.5 LTS (GNU/Linux 4.2.0-42-generic x86_64)

 * Documentation:  https://help.ubuntu.com/

381 packages can be updated.
309 updates are security updates.

Last login: Wed Oct 11 22:30:28 2023 from 10.0.2.7
cybersec-client@ubuntu:~$
```

Output for ssh connection after configuring the rules.

```
cybersec-client@ubuntu: ~
cybersec-client@ubuntu:~$ sudo iptables --policy INPUT DROP
cybersec-client@ubuntu:~$ sudo iptables --policy OUTPUT DROP
cybersec-client@ubuntu:~$ sudo iptables --policy FORWARD DROP
cybersec-client@ubuntu:~$ sudo iptables -A INPUT -p tcp --sport 80 -m state --state NEW,ESTABLISHED -j ACCEPT
cybersec-client@ubuntu:~$ sudo iptables -A INPUT -p tcp --sport 443 -m state --state NEW,ESTABLISHED -j ACCEPT
cybersec-client@ubuntu:~$ sudo iptables -L
Chain INPUT (policy DROP)
target     prot opt source               destination
ACCEPT     tcp  --  anywhere             anywhere             tcp spt:http state NEW,ESTABLISHED
ACCEPT     tcp  --  anywhere             anywhere             tcp spt:https state NEW,ESTABLISHED
ACCEPT     tcp  --  anywhere             anywhere             tcp spt:http state NEW,ESTABLISHED
ACCEPT     tcp  --  anywhere             anywhere             tcp spt:https state NEW,ESTABLISHED

Chain FORWARD (policy DROP)
target     prot opt source               destination

Chain OUTPUT (policy DROP)
target     prot opt source               destination
cybersec-client@ubuntu:~$
```

Configuration of iptables on Cybersec-server to permit HTTP and HTTPS request.